

MAHAMMED UMAR FAARUK

AI and Machine Learning Engineer

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PROFESSIONAL SUMMARY

B.Tech. graduate in Computer Science and Engineering (AI and ML) with hands-on experience developing and deploying machine learning models, data pipelines, and cloud-integrated applications. Published researcher in deep learning-based object detection (93.5% accuracy, 60 FPS with YOLOv11). Contributed to India's first regional-language LLM initiative. Proficient in Python, PyTorch, scikit-learn, NumPy, Pandas, and AWS. Seeking to apply production ML engineering skills to supply chain and logistics AI at Kaleris.

EXPERIENCE

AI Developer Intern | *Viswam AI (Swecha, IIIT Hyderabad and Meta)* May 2025 - Jul 2025 | Hyderabad, India

- Contributed to India's first Telugu Large Language Model (LLM) initiative, gaining direct exposure to NLP data pipelines, prompt engineering, and LLM fine-tuning workflows for regional language AI preservation.
- Built and deployed Tune Buddy AI Chatbot by integrating an LLM inference API (Dify.ai) with a Streamlit front-end, managing end-to-end data flow from user input to personalised model response.
- Architected Craft Connect on AWS (ap-south-1) and Render; designed a secure FastAPI and HTTPX API gateway ensuring no credentials or user data were exposed client-side, applying production MLOps service-hardening practices.
- Implemented an offline-sync mechanism with automatic cloud synchronisation to Supabase PostgreSQL, improving platform accessibility in low-connectivity environments.

Web Development Intern | *Intrainz*

Oct 2024 - Dec 2024 | Remote

- Designed and delivered three functional JavaScript web applications including a calculator, e-commerce storefront, and task manager, applying DOM manipulation, state management, and mobile-first responsive design principles.
- Built a personal portfolio site from scratch using semantic HTML, CSS, and JavaScript, demonstrating end-to-end project delivery and technical documentation skills.

PROJECTS

Fire and Smoke Detection System (Published Research)

Tools: PyTorch, YOLOv11, Python, OpenCV, Twilio SMS API

- Designed and trained a real-time computer vision pipeline for fire and smoke detection achieving 93.5% mAP accuracy at 60 FPS, suitable for production safety-monitoring environments.
- Built a complete data pipeline covering ingestion, preprocessing, augmentation, labelling, and model evaluation; benchmarked YOLOv11 against multiple baseline algorithms using mAP, F1-Score, Precision, and Recall metrics.
- Integrated automated SMS incident alerting via Twilio API to simulate a real-time model monitoring and incident-response workflow.
- Research accepted for publication in an AI Healthcare book (2025, In Press); demonstrates ability to communicate technical ML results clearly to stakeholders.

Adaptive AI-SIEM - Federated Cyber Threat Detection

Tools: PyTorch, CNN, LSTM, Federated Learning, Django, TF-IDF, Blockchain

- Built a multi-model deep learning system (CNN and LSTM) trained on benchmark datasets NSL-KDD and CICDS2017, achieving 99% accuracy and 0.01 False Positive Rate.
- Integrated Federated Learning and Blockchain for distributed model training across IoT edge nodes without sharing raw data, demonstrating knowledge of privacy-preserving ML architecture.
- Conducted structured comparative evaluation of deep learning models against classical algorithms (SVM, Naive Bayes), producing reproducible experiment benchmarks aligned with MLOps best practices.

Craft Connect - Cloud-Integrated Full-Stack Platform

Tools: Python, FastAPI, Supabase, PostgreSQL, AWS, Streamlit, JavaScript

- Architected and deployed a mobile-first platform on AWS with a FastAPI backend and PostgreSQL via Supabase; integrated Google APIs with an external corpus API for verified contributions.
- Engineered a custom offline-sync mechanism with automatic cloud synchronisation, optimising for low-connectivity environments relevant to logistics field operations.

Chess Buddy - AI-Powered Web Application

Tools: Python, Streamlit, Stockfish Engine

- Integrated the Stockfish chess engine into a Python backend for real-time move computation; designed an interactive Streamlit UI managing board states, rule enforcement, and outcome detection; deployed on Streamlit Community Cloud.

TECHNICAL SKILLS

Languages: Python, SQL (basic)

ML and Data Libraries: scikit-learn, NumPy, Pandas, Matplotlib, PyTorch, CNN, LSTM, Federated Learning, Transfer Learning

ML Domains: Supervised and Unsupervised Learning, Classification, Regression, NLP (TF-IDF, Tokenisation, Word Embeddings), Computer Vision, Object Detection (YOLOv11, OpenCV)

Generative AI and LLMs: Prompt Engineering, LangChain, Dify.ai, LLM fine-tuning workflows

Cloud and Deployment: AWS (ap-south-1), Render, Streamlit Cloud, FastAPI, Docker (familiar)

Databases: PostgreSQL, Supabase, Google Firestore

Dev Practices: Git, REST API design, unit testing fundamentals, technical documentation

EDUCATION

B.Tech., Computer Science and Engineering (AI and ML)

Graduated: July 2025

Sreyas Institute of Engineering and Technology, Hyderabad | CGPA: 6.28

CERTIFICATIONS AND PUBLICATIONS

Certifications

- AWS APAC Solutions Architecture Virtual Experience - Forage (Jan 2025)
- Generative AI Workshop - Growth School (Mar 2026)
- Machine Learning Workshop - Innomatics Research Labs (Nov 2025)
- Goldman Sachs Software Engineering Virtual Experience - Forage (Jul 2025)
- Tata Data Visualization Job Simulation - Forage (Jul 2025)
- Machine Learning Workshop - IEEE Sreyas (Dec 2023)

Publication

- Employing Deep Learning Paradigms for Fire and Smoke Detection - Accepted, AI Health Care book, 2025 (In Press)

Technical Articles

- NumPy Basics: A Practical Guide to Understanding NumPy with Real-World Case Studies (2026)
- Mastering Python Functions (2026)